

Controlled-integrator remeasurement of the three-mode relaxation rates

Numerical audit report

7 September 2026

Verdict

The controlled Cartesian calculation removes every suspected numerical feature of the historical action–angle implementation: the state is double precision, the timestep is fixed, standard mathematical functions are used, and there is no positive-action projection or action floor. Nevertheless, the frozen common-plateau gate fails for both bath conditions at both timesteps. Therefore the accessible data do *not* resolve a unique spectral gap. Under the predeclared interpretation gate, the observable dependence is retained as a property of the accessible dynamics (multiple modes have not separated by the latest statistically supported lags), rather than classified as a projection artefact or a timestep-limited result.

This is a negative but sharper conclusion. The resolved action-sector rates are stable under timestep reduction, and the oscillatory mode in $I_1 - I_3$ survives with frequency near the historical value 5.35. What is not supported is replacing the observable-specific decay rates by one common number and calling that number the spectral gap.

1 Frozen design and controlled dynamics

The model has $n = 3$, $\gamma = 0.1$, burn-in 500, stationary measurement time 1000, snapshot interval 0.01, and 64 independent seeded trajectories. Both $(T_1, T_3) = (2, 8)$ and $(5, 5)$ were run at $\Delta t = 10^{-3}$ and 2.5×10^{-4} . The full protocol, including all fit and interpretation gates, was frozen before production.

The controlled state is $c_j = x_j + iy_j$ with $I_j = |c_j|^2$ and $E = H/2$. Interior dynamics is $dc_j = i\nabla_j E dt$; each boundary obeys

$$dc_r = (i - \gamma)\nabla_r E dt + \sqrt{2\gamma T_r} dW_r.$$

Every fixed step uses an implicit-midpoint Hamiltonian update and Cartesian Euler–Maruyama bath updates, with alternating bath order. This is the same projection-free representation used by the validated entropy sampler. The simulated state is float64; the twelve measured observables are stored as float32. The maximum arithmetic discrepancy in the stored sum/difference identities was 8.94×10^{-8} .

2 Frozen plateau results

Figure 1 shows the local five-point log derivatives and trajectory-bootstrap bands. Thick segments mark plateaus accepted by the frozen rule. Table 1 gives every requested observable, including unresolved outcomes.

Table 1: Frozen-rule plateau results for every saved observable.

Case	Observable	Window	$\hat{\lambda}_R$ [95% CI]	N_{eff}
driven, 10^{-3}	$\cos \theta_1$	–	UNRESOLVED	108664
driven, 10^{-3}	$\sin \theta_1$	–	UNRESOLVED	279157
driven, 10^{-3}	$\cos \theta_3$	[1.50, 2.51]	-0.9080 [-1.1010, -0.7094]	121229
driven, 10^{-3}	$\sin \theta_3$	–	UNRESOLVED	376027
driven, 10^{-3}	$\cos(\theta_1 - \theta_3)$	–	UNRESOLVED	235152
driven, 10^{-3}	I_2	[1.00, 2.99]	-0.9440 [-0.9906, -0.8998]	54093
driven, 10^{-3}	$I_1 + I_3$	[1.26, 2.51]	-0.9491 [-1.0380, -0.8622]	73980
driven, 10^{-3}	$I_1 - I_3$	–	UNRESOLVED	173306
driven, 10^{-3}	$\cos \theta_1 + \cos \theta_3$	[1.00, 1.88]	-1.1878 [-1.2911, -1.0932]	79710

Case	Observable	Window	$\hat{\lambda}_R$ [95% CI]	N_{eff}
driven, 10^{-3}	$\cos \theta_1 - \cos \theta_3$	–	UNRESOLVED	218429
driven, 10^{-3}	$\sin \theta_1 + \sin \theta_3$	–	UNRESOLVED	395151
driven, 10^{-3}	$\sin \theta_1 - \sin \theta_3$	–	UNRESOLVED	248210
driven, 2.5×10^{-4}	$\cos \theta_1$	–	UNRESOLVED	105896
driven, 2.5×10^{-4}	$\sin \theta_1$	–	UNRESOLVED	278826
driven, 2.5×10^{-4}	$\cos \theta_3$	[1.00,2.11]	-1.0921 [-1.2129,-0.9686]	126945
driven, 2.5×10^{-4}	$\sin \theta_3$	–	UNRESOLVED	376126
driven, 2.5×10^{-4}	$\cos(\theta_1 - \theta_3)$	–	UNRESOLVED	245876
driven, 2.5×10^{-4}	I_2	[1.00,3.35]	-0.9351 [-0.9975,-0.8697]	55731
driven, 2.5×10^{-4}	$I_1 + I_3$	[1.00,2.24]	-1.0028 [-1.0768,-0.9364]	74422
driven, 2.5×10^{-4}	$I_1 - I_3$	–	UNRESOLVED	174139
driven, 2.5×10^{-4}	$\cos \theta_1 + \cos \theta_3$	[1.26,2.11]	-1.1218 [-1.2439,-1.0146]	79169
driven, 2.5×10^{-4}	$\cos \theta_1 - \cos \theta_3$	–	UNRESOLVED	218106
driven, 2.5×10^{-4}	$\sin \theta_1 + \sin \theta_3$	–	UNRESOLVED	394433
driven, 2.5×10^{-4}	$\sin \theta_1 - \sin \theta_3$	–	UNRESOLVED	248685
equal T , 10^{-3}	$\cos \theta_1$	–	UNRESOLVED	122635
equal T , 10^{-3}	$\sin \theta_1$	–	UNRESOLVED	332646
equal T , 10^{-3}	$\cos \theta_3$	–	UNRESOLVED	121227
equal T , 10^{-3}	$\sin \theta_3$	–	UNRESOLVED	331774
equal T , 10^{-3}	$\cos(\theta_1 - \theta_3)$	–	UNRESOLVED	242084
equal T , 10^{-3}	I_2	[1.00,1.88]	-0.9360 [-0.9930,-0.8836]	57028
equal T , 10^{-3}	$I_1 + I_3$	[1.00,2.24]	-1.0841 [-1.1670,-0.9997]	78670
equal T , 10^{-3}	$I_1 - I_3$	–	UNRESOLVED	177461
equal T , 10^{-3}	$\cos \theta_1 + \cos \theta_3$	[1.00,1.58]	-1.2501 [-1.3545,-1.1521]	83954
equal T , 10^{-3}	$\cos \theta_1 - \cos \theta_3$	–	UNRESOLVED	224467
equal T , 10^{-3}	$\sin \theta_1 + \sin \theta_3$	–	UNRESOLVED	403043
equal T , 10^{-3}	$\sin \theta_1 - \sin \theta_3$	–	UNRESOLVED	257101
equal T , 2.5×10^{-4}	$\cos \theta_1$	–	UNRESOLVED	117212
equal T , 2.5×10^{-4}	$\sin \theta_1$	–	UNRESOLVED	333400
equal T , 2.5×10^{-4}	$\cos \theta_3$	[1.58,2.51]	-1.0081 [-1.3236,-0.7251]	119815
equal T , 2.5×10^{-4}	$\sin \theta_3$	–	UNRESOLVED	332753
equal T , 2.5×10^{-4}	$\cos(\theta_1 - \theta_3)$	–	UNRESOLVED	234498
equal T , 2.5×10^{-4}	I_2	[1.00,3.35]	-0.8627 [-0.9165,-0.8119]	53438
equal T , 2.5×10^{-4}	$I_1 + I_3$	[1.00,1.88]	-1.0010 [-1.0629,-0.9466]	75878
equal T , 2.5×10^{-4}	$I_1 - I_3$	–	UNRESOLVED	177353
equal T , 2.5×10^{-4}	$\cos \theta_1 + \cos \theta_3$	–	UNRESOLVED	80196
equal T , 2.5×10^{-4}	$\cos \theta_1 - \cos \theta_3$	–	UNRESOLVED	224573
equal T , 2.5×10^{-4}	$\sin \theta_1 + \sin \theta_3$	–	UNRESOLVED	403883
equal T , 2.5×10^{-4}	$\sin \theta_1 - \sin \theta_3$	–	UNRESOLVED	256738

Table 3: Historical versus controlled rates. Blank numerical entries mean the observable was unresolved; † means it was not saved historically.

Bath	Observable	Historical	Controlled: 10^{-3} ; 2.5×10^{-4}
driven	$\cos \theta_1$	UNRESOLVED	UNRESOLVED; UNRESOLVED
driven	$\sin \theta_1$	UNRESOLVED	UNRESOLVED; UNRESOLVED
driven	$\cos \theta_3$	UNRESOLVED	-0.9080 [-1.1010,-0.7094]; -1.0921 [-1.2129,-0.9686]
driven	$\sin \theta_3$	not saved†	UNRESOLVED; UNRESOLVED
driven	$\cos(\theta_1 - \theta_3)$	UNRESOLVED	UNRESOLVED; UNRESOLVED
driven	I_2	-0.9926 [-1.0517,-0.9337]	-0.9440 [-0.9906,-0.8998]; -0.9351 [-0.9975,-0.8697]
driven	$I_1 + I_3$	-1.1088 [-1.2013,-1.0228]	-0.9491 [-1.0380,-0.8622]; -1.0028 [-1.0768,-0.9364]

Bath	Observable	Historical	Controlled: 10^{-3} ; 2.5×10^{-4}
driven	$I_1 - I_3$	UNRESOLVED	UNRESOLVED; UNRESOLVED
equilibrium	$\cos \theta_1$	-1.1878 [-1.3607,-1.0160]	UNRESOLVED; UNRESOLVED
equilibrium	$\sin \theta_1$	UNRESOLVED	UNRESOLVED; UNRESOLVED
equilibrium	$\cos \theta_3$	-1.1334 [-1.3366,-0.9465]	UNRESOLVED; -1.0081 [-1.3236,-0.7251]
equilibrium	$\sin \theta_3$	not saved [†]	UNRESOLVED; UNRESOLVED
equilibrium	$\cos(\theta_1 - \theta_3)$	UNRESOLVED	UNRESOLVED; UNRESOLVED
equilibrium	I_2	-0.9065 [-0.9665,-0.8442]	-0.9360 [-0.9930,-0.8836]; -0.8627 [-0.9165,-0.8119]
equilibrium	$I_1 + I_3$	UNRESOLVED	-1.0841 [-1.1670,-0.9997]; -1.0010 [-1.0629,-0.9466]
equilibrium	$I_1 - I_3$	UNRESOLVED	UNRESOLVED; UNRESOLVED

Table 4: Predeclared early-window damped fits, all using $t \in [0.05, 1.00]$.

Case	Observable	λ_R [95% CI]	λ_I	95% CI	Period	ΔAIC
driven, 10^{-3}	$I_1 - I_3$	-5.0244 [-5.2494,-4.8249]	5.4009	[5.2495,5.5404]	1.163	-522.3
driven, 10^{-3}	$\cos \theta_1 - \cos \theta_3$	-6.6106 [-7.0738,-6.1611]	3.6464	[3.4687,3.7771]	1.723	-299.1
driven, 10^{-3}	$\sin \theta_1 - \sin \theta_3$	-3.3903 [-3.4321,-3.3471]	4.0007	[3.9091,4.0916]	1.571	-728.9
driven, 2.5×10^{-4}	$I_1 - I_3$	-5.0030 [-5.2637,-4.7820]	5.4997	[5.3420,5.6486]	1.142	-515.6
driven, 2.5×10^{-4}	$\cos \theta_1 - \cos \theta_3$	-6.3392 [-6.9625,-5.8337]	3.3184	[3.0874,3.4924]	1.893	-327.1
driven, 2.5×10^{-4}	$\sin \theta_1 - \sin \theta_3$	-3.3971 [-3.4383,-3.3584]	4.0075	[3.9186,4.0991]	1.568	-689.5
equal T , 10^{-3}	$I_1 - I_3$	-4.1409 [-4.2475,-4.0436]	5.0810	[4.9343,5.2284]	1.237	-688.1
equal T , 10^{-3}	$\cos \theta_1 - \cos \theta_3$	-5.4500 [-5.7716,-5.1626]	3.4098	[3.1736,3.6265]	1.843	-390.0
equal T , 10^{-3}	$\sin \theta_1 - \sin \theta_3$	-3.5275 [-3.5792,-3.4764]	4.3691	[4.2719,4.4675]	1.438	-703.7
equal T , 2.5×10^{-4}	$I_1 - I_3$	-4.1845 [-4.3260,-4.0580]	5.2637	[5.1306,5.3869]	1.194	-878.1
equal T , 2.5×10^{-4}	$\cos \theta_1 - \cos \theta_3$	-5.5695 [-5.9128,-5.2754]	3.4752	[3.2296,3.6941]	1.808	-344.5
equal T , 2.5×10^{-4}	$\sin \theta_1 - \sin \theta_3$	-3.5204 [-3.5634,-3.4736]	4.4719	[4.3728,4.5701]	1.405	-709.4

The most favorable pooled case is the driven fine-timestep run: the four resolved rates span -1.122 to -0.935 , which passes the 20% spread test, but their 95% intervals have no common intersection. The coarse driven run fails both spread and CI intersection. At equal temperature the pooled and exact exchange-even sector tests also fail. No exchange-odd observable has a late positive-correlation plateau under the frozen $t \geq 1$ rule.

3 Timestep control and comparison with the inherited integrator

Where both controlled timesteps resolve an observable, the estimates are timestep-consistent by the predeclared CI-overlap plus 20% rule. In the driven case the relative changes are 18.4% for $\cos \theta_3$, 0.94% for I_2 , and 5.51% for $I_1 + I_3$. At equal temperature they are 8.15% for I_2 and 7.97% for $I_1 + I_3$. Thus there is no pattern in which the common plateau appears only at the finer step; the result is not classified as timestep-limited.

The action-sector values remain close to the inherited calculation. For example, driven I_2 changes from -0.993 historically to -0.944 and -0.935 , while driven $I_1 + I_3$ changes from -1.109 to -0.949 and -1.003 . Their uncertainty intervals overlap or nearly overlap. Removing the projection therefore changes details but does not create a common observable-independent asymptotic window.

4 Exchange combinations and oscillatory modes

The exact endpoint-exchange combinations could not be formed in the historical run because $\sin \theta_3$ was absent. Here both sine endpoints were saved. At equilibrium, sums are exchange-even and differences are exchange-odd; in the driven case these names only identify the same coordinates because the baths break exchange symmetry.

All predeclared early-window damped fits strongly beat the nested real exponential (Table 4; every ΔAIC is below -299), and all 2000 bootstrap replicates succeeded. The $I_1 - I_3$ frequencies are 5.401 and 5.500 in the driven runs and 5.081 and 5.264 in equilibrium. Their intervals are compatible across the two timesteps and with the historical estimate 5.35 (period 1.174). Hence that particular complex-mode frequency is confirmed.

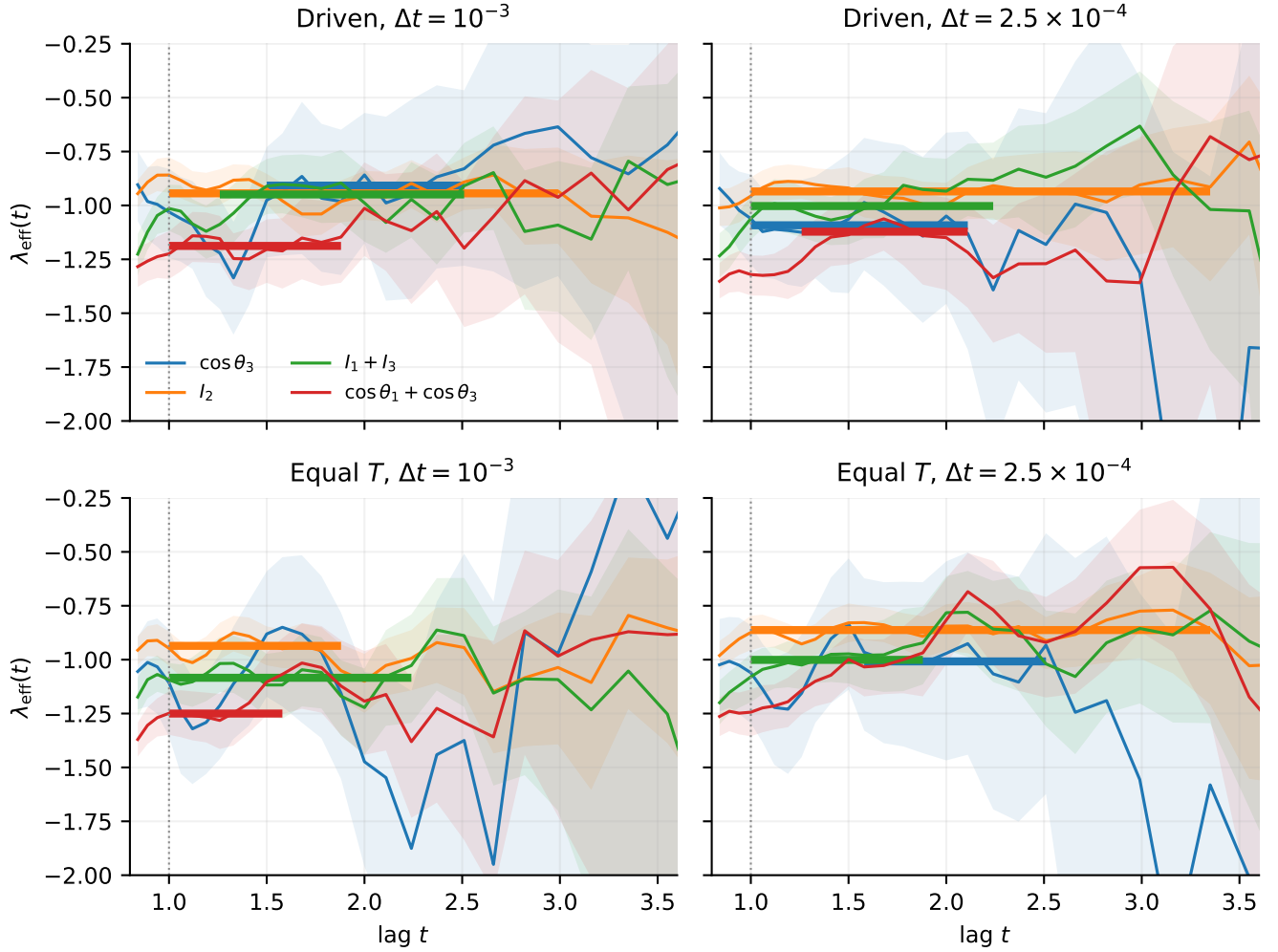


Figure 1: Controlled local decay rates. Bands are trajectory-bootstrap 95% intervals; thick horizontal segments are the weighted rates on intervals accepted by the unchanged plateau rule. Loss of support at later lags produces the visible noisy excursions and is not used to select a rate.

The other exact odd combinations do not share the same frequency: $\cos \theta_1 - \cos \theta_3$ gives approximately 3.3–3.6 and $\sin \theta_1 - \sin \theta_3$ gives approximately 4.0–4.5. Thus the odd sector contains more than one visible damped mode in the predeclared early window; the value 5.35 is not a universal odd-sector eigenfrequency.

5 Numerical integrity, burn-in, and effective support

Each run contains exactly 64 streams and 100001 snapshots per stream. Every saved value is finite. Across all four runs the projection count, floor count, zero-radius count, midpoint-failure count, and nonfinite count are all zero. The minimum sampled actions range from 1.17×10^{-8} to 1.64×10^{-7} , so the controlled calculation directly samples the near-zero-action region without clipping it.

The unchanged discard-first-quarter burn-in metric passes for every observable in every run; its worst value is 0.00498, an order of magnitude below the 0.05 gate. The smallest effective count at the terminal lag of any accepted plateau is 53259. Statistical support is therefore substantial, although it still ends before a common asymptotic mode separates.

Table 2: Pooled common-plateau gate. The eligible column lists observables that individually passed the frozen plateau rule.

Case	Eligible	Spread pass	CI intersection	Verdict
driven, 10^{-3}	$\cos \theta_3, I_2, I_1 + I_3, \cos \theta_1 + \cos \theta_3$	no	no	NO COMMON PLATEAU
driven, 2.5×10^{-4}	$\cos \theta_3, I_2, I_1 + I_3, \cos \theta_1 + \cos \theta_3$	yes	no	NO COMMON PLATEAU
equal T , 10^{-3}	$I_2, I_1 + I_3, \cos \theta_1 + \cos \theta_3$	no	no	NO COMMON PLATEAU
equal T , 2.5×10^{-4}	$\cos \theta_3, I_2, I_1 + I_3$	yes	no	NO COMMON PLATEAU

Table 5: Run-level numerical and stationarity audit. $N_{\text{eff, last}}$ is the smallest effective count among accepted plateaus after accounting for the terminal plateau lag.

Case	Proj./floor	Midpoint fail	Nonfinite	Worst burn-in metric	$N_{\text{eff, last}}$
driven, 10^{-3}	0/0	0	0	0.00498	53931
driven, 2.5×10^{-4}	0/0	0	0	0.00218	55544
equal T , 10^{-3}	0/0	0	0	0.00180	56921
equal T , 2.5×10^{-4}	0/0	0	0	0.00310	53259

6 Provenance and reproducibility

The frozen remote commit is `fddd8c5a1700675a702aadab5dac9262ab48f6c6`; the equivalent local launch commit is `899a5d345a98e4c8d5605a6a5a166a08bb86adcd`. The production source SHA-256 is `58c871882f1180701a7aae637a8f4be113315a72fb9d7815562f5fc90faf7976`; the binary SHA-256 is `f7cd820bb8f716a8790f688db060259da5657e4f20fa09100546e66efb696b3f`. Production ran from 17:45:09 to 17:47:23 UTC on 7 September 2026.

The base seeds are 2026090801, 2026090802, 2026090803, and 2026090804 in the matrix order stated above; all 64 realized stream seeds are recorded in each `metadata.csv`. Exact commands are recorded in `provenance/production_commands.txt`. The integrity audit and SHA-256 of every raw stream are in `provenance/integrity_audit.json` and `provenance/raw_artifact_manifest.sha256`.

Claim boundary

Supported: projection-free, fixed-step controlled dynamics yields stable observable-specific decay rates; the driven action observables decay near -0.94 to -1.00 in the accepted windows; and the $I_1 - I_3$ complex mode has frequency near 5.35. Not supported: a common late-time rate across the requested observables, a unique numerical spectral gap, or identification of all exchange-odd probes with a single complex eigenpair.

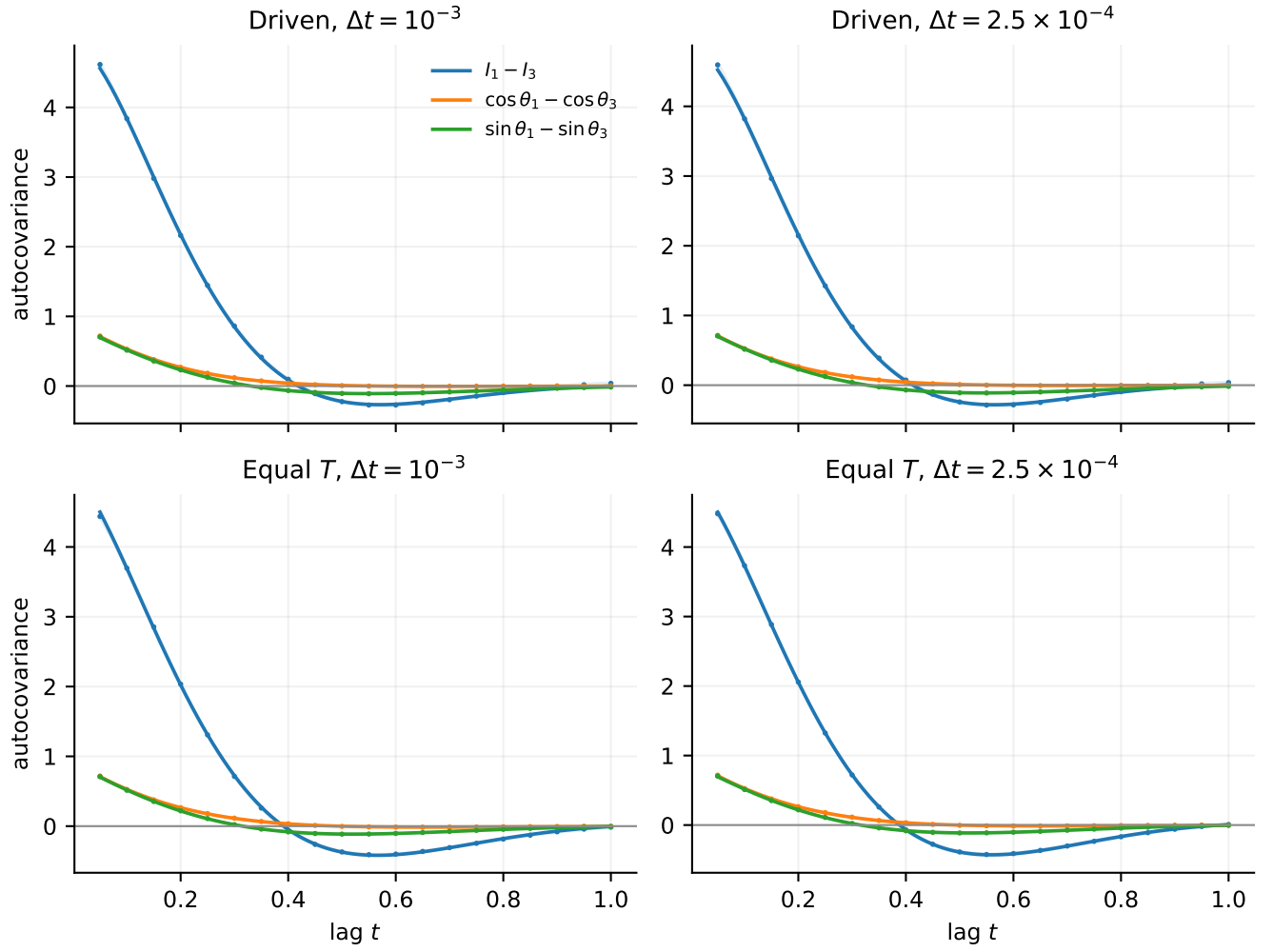


Figure 2: Direct autocovariances (points and one-standard-error bands) and predeclared damped-oscillation fits (lines) on $t \in [0.05, 1.00]$.